

Quiz 02. Exact, linear, and Bernoulli first-order ODEs
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Instructions. In the following problems, the character d is reserved for derivatives and differentials so that, for instance, $\frac{d\star}{d\omega}$ reads “the derivative of $\star$ with respect to ω ” and $d\alpha$ is “the differential of α ”. Per common convention, e is the base of the natural logarithm, and $\exp(abc) = e^{abc}$. Each item in each Scenario is worth two (2) points.

Scenario 1. Find the solution $u(x, y) = C$ to

$$\left(1 + \frac{2}{y}\right) \sin t \, \dot{y} + \cos t = 0$$

by finding an integrating factor μ as a function of y alone.

1. Compute μ when $y = 1.7$.
2. Compute u when $x = \frac{\pi}{3}$ and $y = 1.7$.
3. If $y(0.5\pi) = 1$, determine C .
4. If $y(0.5\pi) = -1$, determine C .
5. Is the equation linear in y ?
6. Is the equation variable-separable?
7. Is the equation homogeneous¹?

Scenario 2. Find the solution $u(x, y) = C$ to

$$(x^2 + y^2 - 5) \, dx = (y + xy) \, dy$$

by finding an integrating factor μ as a function of x alone.

1. Compute μ when $x = 3$.
2. Compute u when $x = e - 1$ and $y = 1$.
3. If $y(0) = 1$, determine C .
4. If $y(0) = -1$, determine C .
5. Is the equation linear in y ?
6. Is the equation variable-separable?
7. Is the equation homogeneous?

Scenario 3. Find the solution $u(x, y) = C$ to

$$(3x^2y + 4xy^2) \, y' = -4xy^2 - 3y^3$$

by finding an integrating factor μ as a function of the form $x^a y^b$, where a and b are real numbers.

1. Determine a .
2. Determine b .
3. Compute μ when $x = -1.7$ and $y = 2.1$.
4. Compute u when $x = -1.7$ and $y = 2.1$.
5. If $y(-1.7) = 3$, determine C .
6. If $y(-1) = 0.5$, determine C .
7. Is the equation linear in y ?
8. Is the equation variable-separable?
9. Is the equation homogeneous?

Scenario 4. Find the solution $x(t)$ to

$$x - (1 + t^2)^{-1.5} + x\dot{x} = 0$$

1. Is the equation exact?
2. Is the equation variable-separable?
3. Is the equation linear?
4. If $x(1) = 1$, determine $x(1.7)$.
5. If $x(4) = 2.1$, determine $x(1)$.
6. If $x(2) = 1.4$, determine $x(3.17)$.

Scenario 5. Some higher-order equations can be tackled as a first-order linear equation by some clever substitution of variables. One example is when the “zeroth derivative” of the dependent variable is missing from the equation. For such a case, one can define a “temporary dependent variable” u equal to the first derivative of the original dependent variable. Use that technique to find the solution $s(t)$ to

$$t\ddot{s} - 2\dot{s} = t^3 e^t$$

by introducing $u = \dot{s}(t)$. Assume $t \neq 0$. You should arrive at a first-order equation that is linear in u .

1. Is the original equation linear in s ?
2. If $s(1) = e + 4$ and $\dot{s}(1) = e + 3$, determine $s(-1.5)$.
3. If $s(1) = e + 1$ and $s(2) = 2e^2 + 8$, determine $s(-1.5)$.

Scenario 6. The time rate of change of a mass w mg is empirically observed to be $2w + tw^5$, where t is in minutes. Set up the differential equation and solve for $w(t)$.

1. Is the equation linear in w ?
2. If at the start of observation, the mass is 2 mg, what was the mass 30 seconds prior?
3. If at the start of observation, the mass is 1 mg, what was the mass 45 seconds prior?

Marking. The total percentage rating a student gets for this Activity is computed as $0.59(0.9^m r) + 0.41p$, where m is the number of unmet requirements and violated instructions, r is the raw score (in percent), and p is the peer rating (in percent) by his/her groupmates.

Submission. Answers shall be communicated through the accompanying Google Form. Peer ratings shall be communicated via the submission platform in the Google Classroom.

¹ Homogeneity all throughout this activity is not according to linear ODEs whose standard forms have zero right-hand sides. Instead, think of homogeneity of functions. See [Sections 2.2 and 2.3](#) of

[Supplemental Material 01](#). Recall that this is the same semantics used in [Quiz 01](#).